

Edge AI Plant Care Monitor

On-device species ID, health assessment & daily care advice

Arduino UNO Q · ECE 180 · Final Project Proposal

Team #4

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What We Are Proposing

A self-contained device that photographs a plant, identifies its species, assesses its health, and fuses that with live soil sensors to give a **daily care recommendation** — all running on the Arduino UNO Q, no cloud.

Where the AI/ML runs (on the AUQ):

- ▶ Two TFLite CNNs execute **on the UNO Q Ubuntu MPU**: *species ID* and *health classification* (healthy / stressed / diseased).
- ▶ A fusion + rule layer combines model output with soil moisture & pH against per-species ideal ranges to generate the recommendation — also on-device.

Must haves

- ▶ On-device species classification
- ▶ On-device health classification
- ▶ Live soil moisture + pH read on STM32
- ▶ Fused daily care recommendation

Nice to haves

- ▶ Fine-tuning on our own plant photos
- ▶ Trend tracking / history dashboard
- ▶ Live display output (MIPI-DSI)
- ▶ Small on-device LLM for phrasing advice

Project Plan — Gantt Chart (Week 2 → Week 5)

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